

# Exercise Sheet 10

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## Cascades and Temporal Networks

5942UE Network Science

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# 10.A Cascades and Threshold Rules

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# Exercise 10.A.1: Threshold Cascade by Hand

Graph (edges  $A-B$ ,  $B-C$ ,  $A-D$ ,  $C-E$ ,  $D-F$ ) with thresholds:  $A(0.4)$ ,  $B(0.5)$ ,  $C(0.3)$ ,  $D(0.5)$ ,  $E(0.5)$ ,  $F(0.3)$

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Seed at  $t = 0$ :  $\{A, C\}$ . Synchronous update.

1. Compute the degree of each non-seed node.
2. Trace which nodes adopt at  $t = 1, 2, 3$ .
3. Does the cascade reach all nodes?

## Solution:

1. Degrees:  $B=2$ ,  $D=2$ ,  $E=1$ ,  $F=1$ .

2.  $t = 1$ :

- $B$ :  $2/2 = 1.0 \geq 0.5 \rightarrow$  **adopts**
- $D$ :  $1/2 = 0.5 \geq 0.5 \rightarrow$  **adopts**
- $E$ :  $1/1 = 1.0 \geq 0.5 \rightarrow$  **adopts**
- $F$ :  $0/1 = 0.0 < 0.3 \rightarrow$  not yet

$t = 2$ :  $F$ :  $1/1 = 1.0 \geq 0.3 \rightarrow$  **adopts**.

3. Yes — full adoption by  $t = 2$ .  $F$  is last because it required  $D$  to adopt first.

# Exercise 10.A.2: Threshold Heterogeneity

Same graph as 10.A.1. Two experiments:

- **A:** All thresholds raised to 0.6.
- **B:** Only  $B$ 's threshold raised to 0.6 (others as in 10.A.1).

For seed  $\{A, C\}$ :

1. Does the cascade reach all nodes in (A)?
2. Does the cascade reach all nodes in (B)?
3. Why does a single threshold change "block" the cascade in (A)?

## Solution:

1. **Experiment A:**  $D$  has  $1/2 = 0.5 < 0.6 \rightarrow$  **does not adopt**.  $F$  depends on  $D$ , also blocked. Cascade fails to reach  $\{D, F\}$ .
2. **Experiment B:**  $B$  still has  $2/2 = 1.0 \geq 0.6 \rightarrow$  adopts. All other thresholds unchanged  $\rightarrow$  full cascade reaches all nodes.
3. In (A),  $D$  requires both neighbours adopted.  $D$ 's only neighbours are  $A$  (seed) and  $F$ .  $F$  won't adopt until  $D$  does — circular dependency. **Dense local clusters help complex cascades; isolated low-degree nodes with high thresholds block spread.**

# Exercise 10.A.3: Cascade Simulation

Simulate threshold cascades on `nx.barabasi_albert_graph(100, 3)`.

1. At what seed size does the cascade reliably sweep most of the network?
2. Raise threshold to 0.5 — how does final adoption change?
3. Compare seeding **hubs** (top- $k$  degree) vs **random**.

## Solution:

```
import networkx as nx
import random

def threshold_cascade(G, seeds, thresholds):
    adopted = set(seeds)
    changed = True
    while changed:
        changed = False
        for node in G.nodes():
            if node in adopted:
                continue
            nbrs = list(G.neighbors(node))
            if not nbrs:
                continue
            frac = sum(1 for n in nbrs if n in adopted) / len(nbrs)
            if frac >= thresholds[node]:
                adopted.add(node)
                changed = True
    return adopted
```

With  $\theta = 0.3$ , even 3–5 seeds can trigger a global cascade thanks to hubs. At  $\theta = 0.5$  random cascades almost always fail — crossing 50% requires dense local reinforcement. **Hub seeding** is much more effective than random because hubs apply pressure to many neighbours at once.



# Exercise 10.A.4: Bridges Differ for Simple vs. Complex Contagion

Dumbbell: two 5-cliques  $\{1..5\}$  and  $\{6..10\}$  joined by bridge 5–6. Node 1 adopts first.

1. **Simple contagion** ( $\beta = 1$ ): does it spread to clique B?
2. **Complex contagion** ( $\theta = 0.4$ ): does it cross the bridge?
3. What change would let the cascade cross?

## Solution:

1. Yes — once node 5 is infected (quickly within clique A), the infection crosses to node 6 and rapidly fills clique B. Bridges are **superhighways** for simple contagion.
2. Within clique A nodes have degree 4; once 2 neighbours adopt ( $50\% > 40\%$ ) adoption spreads. But node 6 has degree 4, with **only** its bridge neighbour (5) adopted:  $1/4 = 25\% < 40\%$ . **Cascade does not cross.**
3. Add a second bridge (e.g., 4–7); or lower  $\theta$  for bridge-adjacent nodes; or plant a seed in clique B. Bridges that help simple contagion *hinder* complex contagion by providing too little adoption pressure on the far side.

# 10.B Temporal Networks

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# Exercise 10.B.1: Order-Dependent Reachability

Three nodes  $A, B, C$  with two timed edges:

1. Can information travel  $A \rightarrow C$ ? Trace the path.
  2. Can it travel  $C \rightarrow A$ ?
- $A-B$  exists at  $t = 1$  only
  - $B-C$  exists at  $t = 2$  only
3. Reverse the timing ( $A-B$  at  $t = 2$ ,  $B-C$  at  $t = 1$ ). Does  $A \rightarrow C$  still work?
  4. What does this say about temporal vs. static reachability?

## Solution:

1. Yes.  $A \xrightarrow{t=1} B \xrightarrow{t=2} C$ .
2. No. At  $t = 1$  the  $B-C$  edge does not yet exist; at  $t = 2$  the  $A-B$  edge no longer exists. No time-respecting path.
3. No —  $B-C$  at  $t = 1$  comes too early ( $A$  has no link yet);  $A-B$  at  $t = 2$  comes too late.
4. Temporal reachability is **asymmetric and order-dependent** even with undirected edges. The same nodes and edges can support completely different information flows depending on timing.

# Exercise 10.B.2: Static Aggregation Hides Causal Order

A hospital contact network over 3 days:

| Day | Edge            |
|-----|-----------------|
| 1   | Nurse–Patient A |
| 2   | Nurse–Patient B |
| 3   | Doctor–Nurse    |

1. Can infection spread from Patient A to Doctor? Trace it.
2. Can it spread from Doctor to Patient A?
3. What might static contact tracing wrongly conclude?

## Solution:

1. Yes. PatientA  $\xrightarrow{d=1}$  Nurse  $\xrightarrow{d=3}$  Doctor.
2. No. Doctor only contacts Nurse on day 3 — both patient contacts happened on days 1 and 2, before the doctor was exposed.
3. A static contact-tracing system sees Doctor–Nurse–PatientA–PatientB as one connected component and would suggest **bidirectional risk**. The real risk is unidirectional (patients  $\rightarrow$  nurse  $\rightarrow$  doctor). Wrong people could be quarantined while real exposure direction is missed.

# Exercise 10.B.3: Temporal Reachability in Code

Implement BFS over a temporal edge list.

1. From node 0, which nodes are reachable and at what earliest time?
2. From node 3, is node 0 reachable?
3. Add edge  $(3, 0, 5)$ . How does reachability change?



## Solution:

```
from collections import deque

def temporal_reachable(edges, source, t_start=0):
    """BFS over (u, v, t) edges; an edge is traversable iff
       t >= current arrival time at u."""
    arrival = {source: t_start}
    queue = deque([(t_start, source)])
    while queue:
        t_curr, node = queue.popleft()
        for (u, v, t) in edges:
            if t < t_curr:
                continue
            nb = v if u == node else u if v == node else None
            if nb is not None and nb not in arrival:
                arrival[nb] = t
                queue.append((t, nb))
    return arrival

temporal_reachable(edges = [(0, 1, 1), (1, 2, 2), (2, 3, 3), (3, 0, 4), (0, 2, 4)],
```

Adding a late edge  $(3, 0, 5)$  provides no new reachability if the nodes already met at an earlier time, but it would create new  $3 \rightarrow 0$  reachability if no earlier time-respecting path existed. Reversing time would break the forward chain entirely.

# Exercise 10.B.4: When Timing Kills Spreading

Star with hub  $H$  and leaves  $A, B, C, D$ . Disease starts at  $A$  at  $t = 0$ .

- **Scenario X:** all edges active at  $t = 1$
  - **Scenario Y:**  $H-A$  at  $t = 1$ ,  $H-B$  at  $t = 2$ ,  $H-C$  at  $t = 3$ ,  $H-D$  at  $t = 4$
1. Which nodes get infected in (X)?
  2. In (Y)?
  3. Now suppose the disease has a **1-step infectious window**. In (Y), does it spread beyond  $H$ ?
  4. What does this illustrate about epidemic containment?

## Solution:

1. All nodes infected at  $t = 1$  — simultaneous contact via  $H$ .
2.  $A \rightarrow H$  at  $t = 1$ ,  $H \rightarrow B$  at  $t = 2$ , then  $C$  at  $t = 3$ ,  $D$  at  $t = 4$ . Eventually all infected.
3. With a 1-step window,  $H$  is only infectious at  $t = 1$ . The  $H-B$  edge appears at  $t = 2$ , after  $H$  has stopped being infectious.  $B, C, D$  are never infected.
4. Even in a fully connected star, **temporal mis-alignment between contact timing and infectious window** can stop spread cold. This is the basis for **contact scheduling** (e.g., staggered shifts in hospitals) as an infection-control strategy.

# Wrap-Up

- threshold cascades depend on dense local reinforcement, not just connectivity
- bridges are good for simple contagion but bad for complex cascades
- temporal networks make reachability asymmetric and order-dependent
- contact timing relative to infectious windows can decisively contain spread